Inside the Earth | Poster presentation script

Suggested delivery: approximately 6–7 minutes including pauses to point to the figures. This is a rehearsal estimate, not a CityU course rule. Use the numbered poster panels as cues; the sample pages do not prescribe a duration.

Opening | four connected views

Hello. We are the DSC5002 project team at City University of Hong Kong. Inside the Earth asks how earthquake patterns change when we look at space, time, depth and magnitude together. Please read the four columns from left to right. We use 46,463 unique USGS earthquake records from 2000 to 2025, all with preferred magnitude at least five. We aim to describe recorded patterns with explicit reference points, not to predict future earthquakes.

Column 1 | space and reference

The equal-area world map shows geographic concentration. But a map alone does not say how unusual proximity to boundaries is relative to available surface area. We therefore generated 65,536 equal-area reference locations on the whole globe, including oceans, and calculated the same spherical distance to PB2002 boundaries. Within one hundred kilometres, the observed share is 70.15 percent, compared with 9.42 percent for the geographic reference. Their ratio is 7.45. A second reference preserves the observed latitude distribution while randomizing longitude; its share is 10.50 percent. These controls help interpret concentration, but neither is a realistic model of earthquake occurrence. We make no causal or significance claim from the ratio.

Column 2 | time and comparability

The second column combines trends with reasons for caution. Recorded activity fluctuates, with 2,701 events in 2011. Comparing the later thirteen years with the earlier thirteen years, the exposure-adjusted rate is 6.63 percent lower at magnitude five and 15.42 percent lower at magnitude six. The regional chart shows that the direction differs: Indonesia and the Philippines decrease by 25.58 percent, whereas the Mediterranean and Himalaya group increases by 8.82 percent. The next chart shows why those numbers should not stand alone. Mw-family composition changes from 73.45 to 55.05 percent. Horizontal-error availability changes sharply too. Removing fixed windows around the largest events reduces the M5 contrast to minus 3.37 percent; this is a sensitivity exercise, not declustering.

Column 3 | depth and relationships

The third column moves below the surface. Most events are shallow, but similar medians hide different tails. Japan and Kuril and Tonga, Kermadec and Fiji both have median depth thirty-three kilometres, yet their deep shares differ, at 5.01 and 11.76 percent. The local cross sections use equal kilometre scales, a fixed swath and zero at the mapped trench. Tonga has 186 deep events out of 640 in the selected swath, compared with 21 out of 351 in northern Tohoku. The correlation heatmap then connects depth to boundary distance across the complete regions. The overall rank correlation is about 0.39 and is stronger in Japan and Kuril. Magnitude correlations are generally weak. These are observational relationships, not causal mechanisms identified by this analysis.

Column 4 | magnitude and event sequences

The last column separates event frequency from an illustrative energy scale. There are only twenty-eight events of magnitude eight or more. They represent 0.06 percent of records but about 71.21 percent of a summed magnitude-based proxy. That proxy increases by about thirty-two for each unit of magnitude. Because preferred magnitude scales are mixed, it is not measured energy in joules and it does not measure damage. Finally, the daily timelines and cumulative curves compare three events within the same five-hundred-kilometre and thirty-day windows. Anchors are excluded. Roughly sixty to seventy-one percent of each post-window sample appears in the first three days. Static aligned panels communicate both the total counts and the timing directly, so an animation link is unnecessary on this poster. The supplementary sensitivity grid shows how counts change when we change the radius and duration.

Conclusion | what the evidence supports

Together, the four columns show strong concentration near mapped boundaries, regional differences in depth, rare high magnitudes and clustered event sequences. They also show how conclusions depend on the reference geography, catalogue composition and selected windows. Our contribution is to make those comparisons visible and reproducible while keeping their limitations explicit. Thank you for your attention. We would be happy to discuss the geographic reference or demonstrate the companion interactive analysis.

Likely questions

Why not 46,523? The supplied all-type extract includes 60 non-earthquake records. The final catalogue is earthquake-only.

Why not fit a predictive b-value? A lower extraction threshold does not establish completeness, and magnitude scales are mixed.

Are the post-window events proven aftershocks? No. Membership is determined only by the stated time and epicentral-distance windows.

What does the colour mean? For location and depth charts: blue shallow, gold intermediate, purple deep. In the timeline only: blue before, gold after; this encoding is stated beside that figure.

Presentation reference

CityU Language Centre: poster script sample. We follow its introduction, signposting, methods, findings and closing structure; the research content and wording here are original.